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On semiabelian groups

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Abstract. Semiabelian groups are a family of finite solvable groups that have played an important role in the inverse Galois problem and the minimal ramification problem. In this paper, we establish several sufficient conditions for a solvable group to be semiabelian mainly by using its Sylow subgroups. We also present a family of non-semiabelian 2-groups.

1 Introduction

Semiabelian groups first appeared within the context of the inverse Galois problem in Saltman [16, Theorem 3.12] and Thompson [20], though they did not name them. Subsequent investigations on the inverse Galois theory of these groups were carried out by Stoll [18] and Dentzer [4]. Moreover, the minimal ramification problem on semiabelian extensions was studied in [12, 13] (see also [15]).

Throughout these studies, properties of semiabelian groups have been gradually uncovered, but there are only a few results regarding which groups belong to the class of semiabelian groups. To fill this gap, the aim of this paper is to give sufficient conditions for a solvable group to be semiabelian. In particular, we note the interplay between the semiabelian and monomial groups. We recall that a monomial group is, by definition, a group whose irreducible complex representations are induced from one-dimensional representations of its subgroups.

Our first result states that the semiabelian property is preserved by isoclinism (see Definition 2.8 for the definition of isoclinism).

Theorem 1.1. *Suppose that G and H are isoclinic finite groups. If G is semiabelian, then H is semiabelian.*

We note that being monomial is also invariant under isoclinism.

Our second result is analogous to the result by Dornhoff on monomial groups (see [5, Theorem 2.6] and also [21, Theorem I.2.10]). His theorem states that if the Sylow p-subgroups of G are modular for all $p \geq 3$ and the Sylow 2-subgroup

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of G is quaternion-free, then G is monomial. For semiabelian groups, we shall prove the following theorem.

Theorem 1.2. *If a finite nilpotent group H is isoclinic to a group G satisfying the following two conditions:*

(P1) *the Sylow p-subgroups of G are modular for all primes $p \geq 3$;*

(P2) *the Sylow 2-subgroup G is quaternion-free, but not isomorphic to a Wilkens group of type C,*

then H is semiabelian.

Wilkens groups of type C appearing in the statement above will be defined in Lemma 3.9 and will be studied in detail in Section 4. Roughly speaking, they are finite 2-groups G fitting into an exact sequence

$$1 \to N \to G \to M_{2^{k+1}} \to 1$$

with some integer $k \geq 3$. Here N is a maximal elementary abelian normal subgroup of G and

$$M_{2^{k+1}} = \langle a, b \mid a^{2^k} = b^2 = 1,\ a^b = a^{1+2^{k-1}} \rangle. \tag{1.1}$$

By noting that being monomial is invariant under isoclinism, if a group satisfies the assumptions in Theorem 1.2, then it is semiabelian and also monomial.

Our results suggest that there is a close relation between semiabelian and monomial groups. We may state the following conjecture.

Conjecture 1.3. *If a group G is semiabelian, then G is monomial.*

A Magma computation shows that there is no semiabelian group among the non-monomial groups of order at most 240.

Moreover, we show the following theorem which supports the conjecture.

Theorem 1.4. *Let $\mathcal{G}$ be an isoclinism invariant class of finite groups with the property that if $G \in \mathcal{G}$, then every subgroup of G is contained in $\mathcal{G}$ and every homomorphic image of G is in $\mathcal{G}$. If $\mathcal{G}$ consists of semiabelian groups, then every $G \in \mathcal{G}$ is monomial.*

The outline of this paper is as follows. In Section 2, we define semiabelian groups and collect several properties of them used in the later sections. Also in this section, we prove the invariance of the semiabelian property under isoclinism (Theorem 1.1). In Section 3, we recall the structure theorems on modular groups and quaternion-free groups. We show that modular p-groups are semi-

abelian (Proposition 3.6). As for 2-groups, we use a classification theorem of non-modular quaternion-free 2-groups due mainly to Wilkens and Janko to obtain a precise result. They classified the non-modular quaternion-free 2-groups into three types: type A, type B, and type C (see Lemma 3.9 for definition). We shall show that, while Wilkens groups of type A and type B are semiabelian (Proposition 3.10), Wilkens groups of type C are *not* semiabelian (Theorem 4.6). A detailed study of Wilkens groups of type C is required to verify this result, which is the focus in Section 4. The type C groups give rise to a family of non-semiabelian groups which are monomial. Using these results, we prove Theorems 1.2 in Section 5. We also give a partial result for general solvable groups in Proposition 5.4. These results are followed by the application of our theorems to solvable isoclinism classes whose stem groups are of order at most 100 (Example 5.5).

Throughout this paper, we assume that all groups are finite. For a finite group G, we denote by G' the derived subgroup of G and by $Z(G)$ the center of G. The Fitting subgroup $F(G)$ is defined as the product of all nilpotent normal subgroups of G and the Frattini subgroup $\Phi(G)$ is the intersection of the maximal subgroups of G. For $x, y \in G$, the commutator is $[x, y] = x^{-1}y^{-1}xy$ and the conjugate x^y is $y^{-1}xy$.

The following standard notation will be also used: C_n is the cyclic group of order n, D_n is the dihedral group of order $2n$, and Q_8 is the quaternion group.

2 Semiabelian groups

Although there are several equivalent definitions of semiabelian group in the literature, we use the following definition which comes from [4, Definition 2.1].

Definition 2.1. A finite group G is called *semiabelian* if it has a sequence

$$G_0 = 1 \leq G_1 \leq \cdots \leq G_n = G$$

such that G_{i+1} is a quotient group of $G_i \ltimes A_i$ with a finite abelian group A_i for all $i = 0, \ldots, n-1$.

It is easy to observe that a semiabelian group is solvable and also that abelian groups are semiabelian.

Lemma 2.2 (Thompson [20] (see also [4, Corollary 2.7])). *We have the following.*

(i) *Nilpotent groups of class* 2 *are semiabelian.*

(ii) *If G is a solvable group whose Sylow p-subgroups are all abelian, then G is semiabelian.*

Although the properties above are easy to prove, difficulties in characterizing semiabelian groups lie in the essentially recursive nature of the definition, which will be seen also in the following lemmas.

The following properties of semiabelian groups will be used in the rest of this paper. The first two lemmas are from Dentzer's paper [4].

Lemma 2.3 ([4, Theorem 2.3]). *A non-trivial finite group G is semiabelian if and only if there are an abelian normal subgroup A and a proper semiabelian subgroup H such that $G = AH$.*

From Lemma 2.3, Dentzer deduces the following.

Lemma 2.4 ([4, Corollary 2.4]). *A non-trivial semiabelian group G has an abelian normal subgroup which is not contained in the Frattini subgroup $\Phi(G)$.*

He also shows the following.

Lemma 2.5 ([4, Theorem 2.8]). *If G and H are semiabelian groups, then the standard wreath product $G \wr H$ is semiabelian. In particular, the direct product $G \times H$ is semiabelian.*

Stoll obtained several useful properties of semiabelian groups in [18].

Lemma 2.6 ([18, Corollary 1 (b)]). *If a group G is an abelian extension of a cyclic group, then G is semiabelian.*

Lemma 2.7 ([18, Lemma 2]). *Every quotient of a semiabelian group is semiabelian.*

Unfortunately, the family of semiabelian groups is not subgroup-closed and thus inductive arguments are sometimes less straightforward.

Using the above lemmas, we shall prove that the semiabelian property is preserved under isoclinism. We first recall the definition of isoclinism.

Definition 2.8 (Hall [6]). The groups G and H are *isoclinic* if an isomorphism $\varphi: G/Z(G) \to H/Z(H)$ induces an isomorphism $\psi: G' \to H'$ satisfying

$$\psi([xZ(G), yZ(G)]) = [\varphi(xZ(G)), \varphi(yZ(G))].$$

If G and H are isoclinic, then we write $H \sim G$ and we call the pair (φ, ψ) an *isoclinism*.

From the definition, for a group G and every abelian group A, the isoclinism $G \sim G \times A$ readily follows.

Now we give a proof of Theorem 1.1.

Proof of Theorem 1.1. Assume $G \sim H$. Let (φ, ψ) be an isoclinism between G and H. If we define

$$C = H \curlywedge G = \{(h, g) \in H \times G \mid \varphi(hZ(G)) = gZ(H)\},$$

then by [11, Theorem 4.2], the group G is a quotient of $H \curlywedge (C/C')$. Hence G is a quotient of $H \times (C/C')$. If H is semiabelian, then $H \times (C/C')$ is semiabelian by Lemma 2.5. It follows from Lemma 2.3 that G is semiabelian. Since isoclinism is an equivalence relation, the reverse implication follows if we exchange the roles of G and H. □

As mentioned before, monomiality is also invariant under isoclinism (see [19, Theorem 3.3]). Furthermore, the nilpotency class and the derived length are also invariant in an isoclinism class.

3 Modular groups and quaternion-free groups

Throughout this section, we denote by p a prime number.

In this section, we study the properties of modular p-groups and quaternion-free 2-groups, from which we deduce sufficient conditions for a p-group to be semiabelian.

We start with the definition of modular groups.

Definition 3.1. A finite p-group G is *modular* if the subgroup lattice of G is a modular lattice, namely, $\langle A, B\rangle \cap C = \langle A, B \cap C\rangle$ for subgroups A, B, C of G such that $A \leq C$.

The following lemma gives a simpler alternative definition.

Lemma 3.2 ([7, Satz 2.2]). *A finite p-group G is modular if and only if $AB = BA$ for all subgroups A, B of G.*

Finite modular groups were first classified by Iwasawa [9] and corrections of his proof were made by Napolitani [14] and Janko [1, Section 9].

Lemma 3.3 ([1, Theorem 9.23]). *A non-Hamiltonian p-group G is modular if and only if it contains an abelian normal subgroup N with cyclic quotient G/N and there exists some $t \in G$ with $G = \langle t, N\rangle$ such that $a^t = a^{1+p^s}$ for all $a \in N$ with a positive integer s. If $p = 2$, then $s \geq 2$.*

Here we recall the definition of Hamiltonian groups.

Definition 3.4. A p-group is called a *Hamiltonian group* if it is non-abelian and all its subgroups are normal subgroups.

It is known that there is no Hamiltonian group if $p \geq 3$ (see [1, Lemma 9.1]). The structure of Hamiltonian 2-groups is well known.

Lemma 3.5 (Dedekind (see [8, III.7.12])). *A Hamiltonian 2-group is isomorphic to a direct product of Q_8 and an elementary abelian 2-group.*

We have the following easy consequence.

Proposition 3.6. *Every modular p-group and every Hamiltonian 2-group is semiabelian.*

Proof. If G is a non-Hamiltonian modular group, then it has an abelian normal subgroup N such that G/N is cyclic by Lemma 3.2. Thus, the result follows from Lemma 2.6.

If G is Hamiltonian, then by Lemma 3.5, we have $G \cong Q_8 \times E$ with E elementary abelian. In particular, as Q_8 is nilpotent of class 2 and E is abelian, G is nilpotent of class 2. Hence G is semiabelian by Lemma 2.2 (i). □

Iwasawa obtained a simple characterization of modular 2-groups. To state it, we need the following definition.

Definition 3.7. A 2-group G is called *quaternion-free* (resp. *D_4-free*) if all homomorphic images of every subgroup of G do not contain a subgroup isomorphic to Q_8 (resp. D_4).

Lemma 3.8 (Iwasawa (see [10, Proposition 1.6])). *A 2-group G is modular if and only if G is D_4-free.*

Some group theorists consider quaternion-free 2-groups as the correct analogue of modular p-groups for $p \geq 3$ (see [5, Section 2]).

Note that a modular and quaternion-free 2-group is clearly semiabelian by Proposition 3.6.

In her paper [22], Wilkens gives a classification of non-modular quaternion-free groups, which is supplemented by Janko [10].

Lemma 3.9 (Wilkens (see [10, Theorem 1.7])). *A 2-group G is non-modular and quaternion-free if and only if it is one of the following three types of groups:*

Type A: *G has a maximal abelian normal subgroup N with exponent larger than 2 and G is a semidirect product $N \rtimes \langle x \rangle$ for some $x \in G$, and if t is an involution in $\langle x \rangle$, then t acts on N by inversion;*

Type B: *there exists a maximal elementary abelian normal subgroup N of G and $x \in G$ such that $G = N\langle x \rangle$, where $\langle x \rangle$ is not normal in G;*

Type C: *there exist a maximal elementary abelian normal subgroup N of G, some $x \in G$, and an involution $t \in G$ with $[N, t] = 1$ such that $G = \langle N, x, t \rangle$. If the order of xN is 2^k, then $G/N \cong M_{2^{k+1}}$ with $k \geq 3$ and $x^{2^k} \neq 1$. Furthermore, we have $[x^{2^{k-1}}, N] = 1$ and $\langle t, x^{2^{k-2}} \rangle \cong D_4$.*

In accordance with [10], we call the groups in Lemma 3.9 *Wilkens groups* of the respective types.

Proposition 3.10. *If a non-modular and quaternion-free group is a Wilkens group of type A or type B, then it is semiabelian.*

Proof. In either case, the quotient group G/N is cyclic, where N is as in Lemma 3.9. Hence the proposition follows from Lemma 2.6. □

In the following section, we will study Wilkens groups of type C more closely.

4 Wilkens groups of type C

Throughout this section, we deal with 2-groups only.

The purpose of this section is to prove that Wilkens groups of type C are *not* semiabelian. To verify this claim, we need to know more properties of the Wilkens groups of type C.

Lemma 4.1 (Janko). *Let $G = \langle N, x, t \rangle$ be a Wilkens group of type C as described in Lemma 3.9. The group G satisfies the following additional properties:*

(i) *N is the unique maximal normal elementary abelian subgroup G;*

(ii) *$\langle t, x^{2^{k-1}} \rangle \cong D_4$;*

(iii) *if we define $\Omega_1(G) = \langle u \in G \mid u^2 = 1 \rangle$, then*

$$\Omega_1(G) = \langle x^{2^{k-1}}, t \rangle N \cong D_4 \times E$$

holds for some elementary abelian 2-group E and $G/\Omega_1(G)$ is a cyclic group of order greater than or equal to 4;

(iv) *subgroups of $\langle x \rangle$ of order greater than 4 are not normal in G.*

This lemma follows from [10, Proposition 1.10] and its proof.

We now collect relations in G.

Lemma 4.2. *Let $G = \langle N, x, t\rangle$ be a Wilkens group of type C as in Lemma 3.9. Assume that the unique normal elementary abelian subgroup N of G is of rank n with basis $a_1, \dots, a_n$. The following relations hold in G:*

$$x^{2^{k+1}} = 1 \qquad \text{with } k \geq 3; \tag{4.1}$$

$$t^2 = 1;$$

$$a_i^2 = 1 \qquad \text{for } i = 1, \dots, n;$$

$$\text{if we set } a = x^{2^{k-1}}, \text{ then } a^t = a^{-1}; \tag{4.2}$$

$$\text{there exists an element } m \in N \text{ such that } x^t = mxa; \tag{4.3}$$

$$[a_i, t] = 1 \qquad \text{for } i = 1, \dots, n;$$

$$[a_i, a] = 1 \qquad \text{for } i = 1, \dots, n;$$

$$[a_i, a_j] = 1 \qquad \text{for } i, j = 1, \dots, n.$$

Moreover, the group G is determined by k in (4.1) and m in (4.3), and the action of x on N.

Proof. Most of these relations are easy consequences of Lemma 3.9. We show (4.2) and (4.3). Since $\langle a, t\rangle \cong D_4$ by Lemma 4.1 (ii), relation (4.2) holds. Also, from $G/N \cong M_{2^{k+1}}$ in Lemma 3.9, we have $(xN)^{tN} = (xN)^{1+2^{k-1}}$ by (1.1), and thus, noting that $[a, N] = 1$, we obtain (4.3).

Since G is an extension of $M_{2^{l+1}}$ by an elementary abelian subgroup N, it is determined by the $M_{2^{l+1}}$-module structure of N with conjugate action. By Lemma 3.9, we have $[N, t] = 1$, and therefore, the module structure is determined by the action of x on N. □

By Lemma 4.2, to determine the structure of G, for a fixed k in (4.1), we have to choose $m \in N$ in (4.3) which is compatible with the action of x on N.

Lemma 4.3. *Let $G = \langle N, x, t\rangle$ be a Wilkens group of type C as in Lemma 3.9. We fix an integer k satisfying (4.1). Let $a = x^{2^{k-1}}$ as in (4.2). Then the element $m \in N$ in (4.3) must satisfy*

$$\prod_{i=1}^{2^{k-1}} m^{x^i} = a^2. \tag{4.4}$$

Proof. We first show that

$$(mxa)^{2^\ell} = x^{2^\ell} \prod_{i=1}^{2^\ell} m^{x^i} \tag{4.5}$$

for every $\ell \geq 2$ by induction on ℓ. Noting that $[a, N] = 1$ and $[a, x] = 1$, we have, for $\ell = 1$,

$$(mxa)^2 = a^2mxmx = a^2mx^2m^x = a^2x^2m^{x^2}m^x = a^2x^2m^xm^{x^2}$$

and, for $\ell = 2$, we obtain

$$\begin{aligned}(mxa)^4 &= a^4x^2m^xm^{x^2}x^2m^xm^{x^2} = x^4(m^xm^{x^2})^{x^2}m^xm^{x^2}\\ &= x^4m^xm^{x^2}m^{x^3}m^{x^4}\end{aligned}$$

since $a^4 = 1$. If equation (4.5) holds for $\ell - 1$, then we have

$$\begin{aligned}(mxa)^{2^\ell} &= ((mxa)^{2^{\ell-1}})^2 = \left(x^{2^{\ell-1}}\prod_{i=1}^{2^{\ell-1}} m^{x^i}\right)^2\\ &= x^{2^\ell}\left(\prod_{i=1}^{2^{\ell-1}} m^{x^i}\right)^{x^{2^{\ell-1}}}\prod_{i=1}^{2^{\ell-1}} m^{x^i} = x^{2^\ell}\prod_{i=1}^{2^\ell} m^{x^i}.\end{aligned}$$

This proves (4.5).

On the other hand, raising both sides of (4.3) to the 2^{k-1}-th power yields

$$a^t = (mxa)^{2^{k-1}}.$$

Applying (4.2) and (4.5) to each side of this equation, we get

$$a^{-1} = x^{2^{k-1}}\prod_{i=1}^{2^{k-1}} m^{x^i},$$

and thus

$$a^2\prod_{i=1}^{2^{k-1}} m^{x^i} = 1.$$

Since order of a is 4 by (4.1), we thus conclude (4.4). □

We next compute the action of x on N.

Lemma 4.4. *Let $G = \langle N, x, t\rangle$ be a Wilkens group of type C as in Lemma 3.9. We fix an integer k satisfying (4.1). Let $Z = \langle x\rangle$. If M is a non-trivial indecomposable Z-submodule of N, then the rank of M is equal to 2^{k-1}.*

Proof. We may assume that N itself is indecomposable. We denote the rank of N by n. Since N is indecomposable, we can choose a basis $(a_1, \dots, a_n)$ of N so that $x \in Z$ acts on N by the Jordan block

$$J_n = \begin{bmatrix} 1 & 1 & & O & \\ & 1 & 1 & & \\ & & \ddots & \ddots & \\ & & & \ddots & 1 \\ O & & & & 1 \end{bmatrix}$$

of size n over the field $\mathbb{F}_2$ of 2-elements. We define $K_n = J_n - E_n$, where E_n is the identity matrix. We then claim that, for all $\ell \geq 2$,

$$\sum_{i=1}^{2^\ell} J_n^i = K_n^{2^\ell - 1} + K_n^{2^\ell}. \tag{4.6}$$

We prove this by induction on ℓ. Here we drop the subscript n for simplicity. We start with the case $\ell = 2$:

$$\begin{aligned} \sum_{i=1}^{2^2} J^i &= (E + K) + (E + K^2) + (E + K + K^2 + K^3) + (E + K^4) \\ &= K^3 + K^4 \end{aligned}$$

by the binomial theorem. If the claim holds for $\ell - 1$, then we compute

$$\begin{aligned} \sum_{i=1}^{2^\ell} J^i &= \sum_{i=1}^{2^{\ell-1}} J^i + \sum_{i=2^{\ell-1}+1}^{2^\ell} J^i = \sum_{i=1}^{2^{\ell-1}} J^i + J^{2^{\ell-1}} \sum_{i=1}^{2^{\ell-1}} J^i \\ &= (E + J)^{2^{\ell-1}} (K^{2^{\ell-1}-1} + K^{2^{\ell-1}}) = K^{2^{\ell-1}} (K^{2^{\ell-1}-1} + K^{2^{\ell-1}}) \\ &= K^{2^\ell - 1} + K^{2^\ell}. \end{aligned}$$

This completes the proof of our claim (4.6).

Now we use the additive notation on N and revive the subscript n. By (4.4), the element a^2 is contained in N and it is left fixed by the action of x, and therefore, we may assume $a_n = a^2$. By (4.6), the condition from Lemma 4.3 amounts to

$$(K_n^{2^{k-1}-1} + K_n^{2^{k-1}})m = a_n.$$

Since K_n is nilpotent, it follows that $n \geq 2^{k-1}$; otherwise, the image is 0. On the other hand, $K_n^{2^{k-1}-1} + K_n^{2^{k-1}}$ sends all a_j with $j > n - 2^{k-1} + 1$ to 0. Hence the image of the map coincides with a_n if and only if $n = 2^{k-1}$ and the a_1-coordinate of m is non-zero. □

We note the following consequence of the above proof.

Corollary 4.5. *Let $G = \langle N, x, t\rangle$ be a Wilkens group of type C. If we take a basis $(a_1, \ldots, a_{2^{k-1}})$ of N so that x acts on N by the Jordan block $J_{2^{k-1}}$, then the a_1 coordinate of m in (4.3) must not be zero.*

We can now state our main theorem in this section.

Theorem 4.6. *If $G = \langle N, x, t\rangle$ is a Wilkens group of type C with a fixed integer k, then G is not a semiabelian group.*

Proof. Let $Z = \langle x\rangle$ as in Lemma 4.4 and $s = 2^{k-1}$.

We first prove the theorem in the case where N is an indecomposable Z-module of rank s. Let $(a_1, \ldots, a_s)$ be a basis of N on which x acts as the Jordan block J_s. Then we have

$$a_1^x = a_1a_2,\ a_2^x = a_2a_3,\ \ldots,\ a_{s-1}^x = a_{s-1}a_s,\ a_s^x = a_s.$$

These equations yield

$$[a_1, x] = a_2,\ \ldots,\ [a_{s-1}, x] = a_s,\ [a_s, x] = 1.$$

Since the Frattini subgroup $\Phi(G)$ is generated by $G^2 = \langle g^2 \mid g \in G\rangle$ and G' by [8, III.3.14], we have $a_2, \ldots, a_s \in \Phi(G)$. By (4.4) with $m = a_1$, we have

$$\prod_{i=1}^{2^{k-1}} a_1^{x^i} = a^2 \in N.$$

Since $\Phi(G)$ is a normal subgroup of G, we see $a_1^{x^i} \in \Phi(G)$ for $i \geq 2$, and thus $a_1^x \in \Phi(G)$. This implies that a_1 is also contained in $\Phi(G)$. Therefore, we have $\langle a\rangle N \leq \Phi(G)$. Now we shall show that $\langle a\rangle N$ is the unique maximal abelian normal subgroup of G. Let A be a maximal abelian normal subgroup of G. Since $\langle a\rangle N$ is a normal abelian subgroup, we have $\langle a\rangle N \leq A$. Lemma 3.9 implies that $G/N \cong M_{2^{k+1}}$, and hence $G/\langle a\rangle N$ is an abelian group generated by the cosets of x and t. Since $[t, a] \neq 1$, we see $t \notin A$. Therefore, if $A > \langle a\rangle N$, then A must contain a power of x whose order is greater than or equal to 8. This is impossible by Lemma 4.1 (iv). We thus conclude $A = \langle a\rangle N \leq \Phi(G)$. It follows from Lemma 2.4 that G cannot be semiabelian.

Now we prove the general case. In this case, we have $N = M \oplus L$, where M is an indecomposable Z-submodule of rank s as in Lemma 4.4 and L is a Z-module. The quotient group G/L is isomorphic to the group considered above, and hence is not semiabelian. By Lemma 2.7, G is not semiabelian, too. □

From the proof above, we obtain the following corollary, which is of its own interest.

Corollary 4.7. *If $G = \langle N, x, t\rangle$ is a Wilkens group of type C, then $\langle x^{2^{k-1}}\rangle N$ is the unique maximal abelian normal subgroup of G.*

Non-abelian groups whose abelian normal subgroups are contained in $\Phi(G)$ are called *irreducible groups* by Dentzer [4] and Schneps [17].

Example 4.8. The smallest example of a Wilkens group of type C is

$$\begin{aligned} G = \langle x, t, a_1, a_2, a_3, a_4 \mid\ & x^{16} = t^2 = a_i^2 = 1,\ (x^4)^t = x^{-4}, \\ & [a_i, t] = [a_i, x^4] = [a_i, a_j] = 1,\ a_1^x = a_1a_2, \\ & a_2^x = a_2a_3,\ a_3^x = a_3a_4,\ a_4^x = x^8\ (i, j = 1, \ldots, 4)\rangle \end{aligned}$$

of order 256. The GAP ID of this group is $(256, 330)$; it is non-semiabelian but monomial.

Let $G_2 = G \times C_2$. The group G_2 is clearly a Wilkens group of type C. We see that $\Phi(G_2) = \Phi(G) \times \Phi(C_2) = \Phi(G)$ and thus G_2 is not irreducible in the above sense. This shows that the reverse implication of Lemma 2.4 is not true in general.

When $k = 4$, the smallest Wilkens group of type C is of order 2^{13}.

5 Proof of Theorem 1.2

Our main aim in this section is to prove Theorem 1.2. We also study the case of general solvable groups. By using Theorem 1.1, Theorem 1.2 clearly follows from the following theorem.

Theorem 5.1. *If a finite nilpotent group G satisfies the following two properties:*

(P1) *the Sylow p-subgroups are modular for all $p \geq 3$;*

(P2) *the Sylow 2-subgroup is quaternion-free, but is not isomorphic to a Wilkens group of type C,*

then G is semiabelian.

Proof. Since G is nilpotent, it is the direct product of its Sylow subgroups. For all $p \geq 3$, the Sylow p-subgroups are semiabelian by Proposition 3.6. For $p = 2$, the 2-Sylow group is semiabelian by Propositions 3.6 and 3.10. Therefore, G is semiabelian by Lemma 2.5. □

Remark 5.2. As in the case of monomial groups [5, Theorem 2.6], one may expect a solvable group satisfying (P1) and (P2) to be semiabelian. In fact, we have not found any counterexample so far.

For general solvable groups, we prove a proposition below. We first recall the definition of Carter subgroups.

Definition 5.3. Let G be a solvable group. A *Carter subgroup* of G is a nilpotent self-normalizing subgroup of G.

Carter showed that such subgroups always exist in solvable groups (see [8, VI.12.2]).

Proposition 5.4. *Assume that G is a solvable group but not a nilpotent group. Let $K_\infty(G)$ be the intersection of the groups of the lower central series of G. If $K_\infty(G)$ is abelian and a Carter subgroup $C(G)$ of G is semiabelian, then G is semiabelian.*

Proof. Since G is not nilpotent, we have $K_\infty(G) \neq 1$. By [8, III.2.5 d)], the quotient $G/K_\infty(G)$ is nilpotent. The subgroup $C(G)K_\infty(G)$ is self-normalizing [3], and thus $C(G)K_\infty(G) = G$ by [8, III.2.3 c)]. Since $C(G)$ is semiabelian by assumption, Lemma 2.3 implies that G is semiabelian. □

Since a Carter subgroup is nilpotent, we can use Theorem 5.1 or Lemma 2.2 to verify that $C(G)$ is semiabelian.

Example 5.5. In this example, we apply our results to solvable isoclinism classes whose stem group is of order less than 100 to show how our results work. Here a group G is called a *stem group* if it satisfies $Z(G) \leq G'$. Recall that every isoclinism class has a stem group as a class representative.

If $|G|$ has no prime divisor ℓ such that $\ell^3 \mid |G|$, then every Sylow p-subgroup is abelian, and thus G is semiabelian by Lemma 2.2. Hence we skip these groups.

The *non*-semiabelian stem groups of order up to 100 are

$$(24,3) = \mathrm{SL}(2,3), \quad (48,28) = C_2.S_4, \quad (64,8) = C_2^2.SD_{16},$$
$$(64,41) = D_8 \rtimes C_4, \quad (96,3) = C_2^2.\mathrm{SL}(2,3), \quad (96,190) = Q_8.D_6,$$
$$(96,201) = Q_8.A_4, \quad (96,203) = D_4.A_4,$$

where the numbers are the GAP IDs of the groups. This list is verified independently of our results.

For the rest of the solvable isoclinism classes, we first check conditions (P1) and (P2) in Theorem 1.2 for nilpotent stem groups. If it fails to satisfy them, then we check the conditions for isoclinic groups up to order 128. For non-nilpotent groups, we first check whether $K_\infty(G)$ is abelian or not, and then the conditions in Proposition 5.4.

Among these classes, the following stem groups are those we *cannot* prove to be semiabelian from our results:

$$\begin{array}{lllll} (24,12), & (32,43), & (54,8), & (64,46), & (64,128), \\ (64,134), & (64,138), & (64,152), & (64,190), & (64,257), \\ (72,15), & (72,43), & (81,7), & (96,64), & (96,115), \\ (96,118), & (96,139), & (96,185), & (96,204), & (96,227). \end{array}$$

In general, our results works well for the above cases. Note that verifying the condition for a stem group implies the semiabelian property of the infinitely many groups in the same isoclinism class. On the other hand, the limit of our results lies partly in the fact that we can only deal with semiabelian groups whose subgroups are also semiabelian. For example, the group $(96,204) = C_2^3 \ltimes A_4$ is semiabelian, but has a subgroup isomorphic to $\mathrm{SL}(2,3)$, which is not semiabelian. As we mentioned before, this fact makes it difficult to characterize semiabelian groups more precisely.

The computation of this example is done by Magma [2].

In closing, we prove Theorem 1.4 supporting Conjecture 1.3.

Proof of Theorem 1.4. Assume that $G \in \mathscr{G}$ is semiabelian. There exists a stem group H such that $H \sim G$, which is also semiabelian by Theorem 1.1. If H is abelian, then G is abelian, and therefore G is monomial. We may assume that H is non-abelian. By Lemma 2.4, there exists an abelian normal subgroup of N of H such that $N \not\subset \Phi(H)$. On the other hand, it follows from [8, III.3.12] that $H' \cap Z(H) \leq \Phi(H)$. Since H is a stem group, this implies $\Phi(H) \geq Z(H)$. Hence we have $N \not\subset Z(H)$. By combining our assumption on $\mathscr{G}$ and [8, V.18.3], this implies that H is monomial. Since $G \sim H$, the group G is also monomial. □

As we have seen in Example 4.8, the converse of Conjecture 1.3 is not valid. It is well known that supersolvable groups are monomial but not always semiabelian. In fact, there is a counterexample of order 192.

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